

Interneuron-Differentiated Inhibition in a Laminar Sparse-Distributed Memory Column: Testing Parvalbumin and Somatostatin Contributions to Encoding and Retrieval

Subham Singh(1506413)

Information Technology, Frankfurt University of Applied Sciences,
Frankfurt, Germany.

1 Background and Motivation

Cortical Learning Algorithms (CLA), including Hierarchical Temporal Memory (HTM) and its associative extension, the Neural Association Algorithm (NAA), model neocortical computation using sparse distributed representations (SDRs), active dendrites, and local unsupervised plasticity. These models learn continually from streaming input without global error signals, which makes them biologically attractive and robust to catastrophic forgetting. Nevertheless, every model in the HTM/CLA/SDR lineage represents inhibition as a single homogeneous mechanism: k-winner-take-all (kWTA) local lateral inhibition that fixes representational sparsity. Biological Cortex does not work in the same way. Inhibition in the neocortex is divided among distinct interneuron types with different targets, kinetics and computational roles- most prominently fast, perisomatic parvalbumin(PV) cells that control gain, timing, and sparsity, and slower, dendrite-targeting somatostatin (SST) cells that gate apical/top-down input and stabilise activity. This cell-type-specific division of labour is well documented experimentally and in detailed spiking column models, but it has never been imported into the discrete-SDR, HTM-style tradition. This thesis investigates whether resolving inhibition into two mathematically distinct interneuron operators within a single laminar SDR column produces a functional dissociation that a homogeneous kWTA cannot, especially the separation between the mechanisms that govern encoding and those that govern retrieval.

2 Problem Statement and Research Gap

Across the three models or works there is a consistent gap:

1. The HTM/CLA/SDR tradition has sparse, continual, unsupervised learning but only one inhibitory abstraction(kWTA), with no cell-type structure.
2. Detailed laminar column models (e.g. V1 models with Parvalbumin(PV), Somatostatin(SST) & Vasoactive Intestinal Peptide Expressing(VIP)) have rich interneuron structure, but are spiking/biophysical and doesn't use SDRs neither frame encoding versus retrieval as the object of study.
3. The PV/SST functional dissociation is established in biology and in rate/attractor models, but has never been tested over discrete SDR assemblies and the encoding vs retrieval angle specifically has not been modelled in this aspect.

No existing model combines (i) a discrete SDR/HTM substrate, (ii) two operator-distinct interneuron types(divisive-normalisation PV, slow dendritic-gate SST) , and (iii) an explicit test of their differential roles in encoding versus retrieval. This is the gap the project addresses.

3 Aim

To design, implement, and evaluate a laminar cortical-column model that replaces HTM's homogenous kWTA inhibition with two distinct interneuron operators, and to test whether PV and SST inhibition contribute differentially to sparse encoding and cue-driven retrieval within an SDR associative-memory column.

4 Research Questions and Hypothesis

This project is guided by four hypothesis:

1. **Hypothesis 1 (Encoding)** - Feedforward input through L4 recruits a sparse L2/3 assembly, and fast PV-mediated perisomatic inhibition (modelled as divisive normalisation) is the principal mechanism controlling assembly size(H1a). PV additionally influences activation- timing precision once a within-step temporal dimension is added (H1b).
2. **Hypothesis 2 (Retrieval)** - A deep-layer/ top-down cue reactivates the learned L2/3 assembly, and SST- mediated dendritic inhibition regulates the spread and persistence of that reactivation.
3. **Hypothesis 3 (Arbitration)**- When basal (feedforward) and apical(retrieval) signals disagree, apical-basal integration produces a graded outcome - blend when consistent, suppression/bias towards the stronger signal when in conflict - rather than a hard binary switch.
4. **Hypothesis 4 (Differential Contribution)** - PV and SST contribute differentially, not exclusively: PV inhibition most strongly affects encoding sparsity, SST inhibition most strongly affects retrieval selectivity and stability, with neither being exactly zero on the other's metric.

Research questions: Can cell- type - differentiated inhibition be introduced into an SDR column while preserving sparse, unsupervised, continual learning? Does it yield an encoding/retrieval dissociation? Is that dissociation functionally distinct from what kWTA can produce?

5 Objectives

1. Implementation of LaminarCorticalUnit - a laminar column (L4 relay - L2/3 assembly with PV and SST operators, L5 retrieval cue; L6 gain-control placeholder) - config- switchable against the existing HTM column.
2. Establish, as the first result, whether the PV operator is functionally distinct from kWTA (the falsification check)
3. Test H1 and H2 via a single- column encode-versus-retrieve simulation (Simulation 1). H1 is evaluated on PV's control of assembly size(H1a) and, following a temporal-dimension extension, on PV's effect on activation-timing precision (H1b); should the temporal extension prove infeasible, H1 falls back to assembly- size control alone and timing is reported as future work.
4. Test H4 via a graded PV/SST strength sweep (Sim 2).
5. Test H3 via a simulation of apical- basal conflict simulation (Sim 3), using a consistency metric(e.g. SDR overlap between basal- and apical- driven candidates) declared in advance.
6. Evaluate all results against the plain HTM column as a baseline, and assess biological plausibility, robustness, and continual- learning behaviour.

6 Methodology

Base Layer : A Rate-coded/ point neuron units over SDRs (no biophysically detailed neurons). Extends the existing HALO corticalunit, following the framework's swappable via config not hard import convention. **Two Distinct inhibitory operators**: Parvalbumin (PV) is a static divisive-normalisation nonlinearity over total drive ($r_i = \frac{d_i}{\varepsilon + \lambda \sum_j d_j}$, threshold) deliberately not smoothed kWTA. SST is a slow leaky-integrator gate applied to the apical stream before it is summed with basal drive, giving it memory/persistence that PV lacks. **The falsification check(central to the project)**. As PV must be shown to differ from HTM's existing kWTA, Simulation 1 will explicitly measure and report whether assembly size varies with total input drive; whether full suppression (an empty assembly) is reachable - a behaviour kWTA, cannot produce. Whether the graded pre-threshold value is retained downstream; and which operating regime the chosen parameters occupy. If PV only ever operates in the high λ "silence everything" regime, that will be reported as not yet demonstrating a functional difference from kWTA. This check runs early, as a go/no-go gate, before the full experimental programme is committed. H1 associates PV with both assembly-size control and activation-timing precision. The current state has no within-step temporal dimension, so a coarse temporal representation(e.g.activation-order or phase) will be added to the L2/3 population to make timing precision measurable(H1b). This keeps the timing claim falsifiable - able to return a genuine positive or negative result. If

the temporal extension proves infeasible within the project timeframe, H1 is evaluated on assembly-size control alone(H1a) and timing is reported as well-defined future direction. The fallback is retained as methodological insurance, not as a reduction in scope.

6.1 Simulations

1. Sim 1 - In this simulation, one column will run separately under an encoding condition(bottom-up via L4 only) and a retrieval condition(top-down cue via L5 only); measure encoding sparsity and retrieval overlap/persistence against the stored pattern. Tests Hypothesis 1 and Hypothesis 2.
2. Sim 2 - A graded sweep over multiple $PV \times SST$ strengths, which will be measured by an encoding metric and a retrieval metric at every combination, where the expected outcome would be that PV will dominate the encoding curve and SST will dominate the retrieval curve without either being exactly zero on the other. This simulation will tests the Hypothesis 4.
3. Sim 3 - Here simultaneous and independently controlled bottom-up and top-down input, implement an apical mismatch gate; which will confirm a graded blend/suppress/bias outcome using a pre-declared consistency metric(e.g. SDR overlap between basal-and apical driven candidates) This simulation tests the hypothesis 3.

7 Data

The project uses two complementary data regimes: controlled synthetic streams to isolate mechanisms with ground-truth labels, and real streams to test robustness and scalability. In the case of Synthetic Streams, they will be custom streams with controlled properties like temporal ambiguity, semantic overlap, and partial observability, which will be specifically generated to isolate specific mechanisms and to provide ground-truth labels for scoring encoding and retrieval directly. These are produced by the HALO contextual-task suite(single-context sequence tasks, interleaved multi-context switch tasks, and random-noise negative controls), which allows the contextual structure to be varied systematically while holding other factors constant.

Apart from the synthetic streams, two real datasets are used to assess robustness and scalability:

1. **Numenta Anomaly Benchmark(NAB)** - The NYC Taxi demand stream (number of NYC taxi passengers in 30-minute buckets, July 2014 - January 2015, 10,320 timestamps, with five labelled anomalous intervals) plus one additional "realKnownCause" stream(e.g. machine temperature or EC2 request-latency system failure). NAB is chosen because it is the HTM-native streaming benchmark and is directly comparable to prior cortical-learning work, making the plain-HTM baseline meaningful.
2. **UCI Human Activity Recognition(HAR)** - multi-context. In this dataset, recordings of 30 subjects performing six activities (walking, walking upstairs, walking downstairs, sitting, standing, lying) from a waist-mounted smartphone's

accelerometer and gyroscope at 50Hz (10,299 instances). The activity labels provide genuine context switches, giving the encoding/retrieval dissociation and specifically SST’s retrieval-stability role, which is a real context-changing stream to be evaluated against, rather than a single-context one.

Pre-processing and Encoding - All the streams are converted to Sparse Distributed Representations via standard HTM-style scalar/category encoders before being presented to the column. Input is delivered as a continuous stream rather than a fixed dataset, so that learning dynamics reflect continual adaptation rather than offline optimisation. No personal or sensitive data are used; both real datasets are public and used under their respective licenses for non-commercial academic research.

8 Novelty and Contribution

The mechanisms this project builds on are all established: the PV-somatic/SST-dendritic division of labour, divisive normalisation as a model of PV inhibition, SST dendritic gating of top-down input, and laminar PV/SST columns. The contribution is therefore one of synthesis and substrate transfer, stated honestly rather than as a new biological mechanism. What is actually unclaimed is the combination of three things: importing two mathematically distinct interneuron operators (divisive-normalisation PV, slow dendritic-gate SST) into HTM/SDR and then using them to test an encoding vs retrieval dissociation over discrete SDR assemblies, an axis not addressed by the spiking predictive-coding column models or the attractor-network tradition (whose dissociation is onset versus termination). i.e. the first-cell-type-resolved inhibition model within the HTM/SDR tradition, showing that two distinct interneuron operators differentially support encoding and retrieval in a non-spiking associative memory column.

9 References

References

- [1] S. Ahmad and J. Hawkins, “Properties of sparse distributed representations and their application to hierarchical temporal memory,” *arXiv preprint arXiv:1503.07469*, 2015.
- [2] Y. Cui, S. Ahmad, and J. Hawkins, “The HTM spatial pooler—a neocortical algorithm for online sparse distributed coding,” *Frontiers in Computational Neuroscience*, vol. 11, art. 111, 2017.
- [3] M. Carandini and D. J. Heeger, “Normalization as a canonical neural computation,” *Nature Reviews Neuroscience*, vol. 13, no. 1, pp. 51–62, 2012.
- [4] N. R. Wilson, C. A. Runyan, F. L. Wang, and M. Sur, “Division and subtraction by distinct cortical inhibitory networks in vivo,” *Nature*, vol. 488, pp. 343–348, 2012.

- [5] L. Hertäg and H. Sprekeler, “Amplifying the redistribution of somato-dendritic inhibition by the interplay of three interneuron types,” *PLoS Computational Biology*, vol. 15, art. e1006999, 2019.
- [6] L. Hertäg and H. Sprekeler, “Learning prediction error neurons in a canonical interneuron circuit,” *eLife*, vol. 9, art. e57541, 2020.
- [7] P. Krishnamurthy, G. Silberberg, and A. Lansner, “A cortical attractor network with Martinotti cells driven by facilitating synapses,” *PLoS ONE*, vol. 7, no. 4, art. e30752, 2012.
- [8] E. Fransén and A. Lansner, “A model of cortical associative memory based on a horizontal network of connected columns,” *Network: Computation in Neural Systems*, vol. 9, no. 2, pp. 235–264, 1998.
- [9] J. Sacramento, R. P. Costa, Y. Bengio, and W. Senn, “Dendritic cortical micro-circuits approximate the backpropagation algorithm,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.
- [10] M. Larkum, “A cellular mechanism for cortical associations: an organizing principle for the cerebral cortex,” *Trends in Neurosciences*, vol. 36, no. 3, pp. 141–151, 2013.
- [11] G. Moreni, C. Dorcioman, C. M. A. Pennartz, and J. F. Mejias, “Cell-type-specific firing patterns in a V1 cortical column model depend on feedforward and feedback-driven states,” *PLoS Computational Biology*, vol. 21, no. 4, art. e1012036, 2025.
- [12] E. L. White, *Cortical Circuits: Synaptic Organization of the Cerebral Cortex—Structure, Function, and Theory*. Boston: Birkhäuser, 1989.
- [13] G. M. Shepherd and S. Grillner (eds.), *Handbook of Brain Microcircuits*, 2nd ed. Oxford University Press, 2018.
- [14] S. M. Sherman and R. W. Guillery, “On the actions that one nerve cell can have on another: distinguishing ‘drivers’ from ‘modulators’,” *Proceedings of the National Academy of Sciences*, vol. 95, no. 12, pp. 7121–7126, 1998.
- [15] A. Lavin and S. Ahmad, “Evaluating real-time anomaly detection algorithms—the Numenta Anomaly Benchmark,” in *IEEE 14th International Conference on Machine Learning and Applications (ICMLA)*, 2015.
- [16] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, “A public domain dataset for human activity recognition using smartphones,” in *European Symposium on Artificial Neural Networks (ESANN)*, 2013.